

Fast algorithm for detecting community structure in networks

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It has been found that many networks display community structure—groups of vertices within which connections are dense but between which they are sparser—and highly sensitive computer algorithms have in recent years been developed for detecting such structure. These algorithms however are computationally demanding, which limits their application to small networks. Here we describe a new algorithm which gives excellent results when tested on both computer-generated and real-world networks and is much faster, typically thousands of times faster than previous algorithms. We give several example applications, including one to a collaboration network of more than 50 000 physicists.

I. INTRODUCTION

There has in recent years been a surge of interest within the physics community in the properties of networks of many kinds, including the Internet, the world wide web, citation networks, transportation networks, software call graphs, email networks, food webs, and social and biochemical networks [1, 2, 3, 4]. One property that has attracted particular attention is that of “community structure”: the vertices in networks are often found to cluster into tightly-knit groups with a high density of within-group edges and a lower density of between-group edges. Girvan and Newman [5, 6] proposed a computer algorithm based on the iterative removal of edges with high “betweenness” scores that appears to identify such structure with some sensitivity, and this algorithm has been employed by a number of authors in the study of such diverse systems as networks of email messages, social networks of animals, collaborations of jazz musicians, metabolic networks, and gene networks [5, 6, 7, 8, 9, 10, 11]. As pointed out by Newman and Girvan [6], the principle disadvantage of their algorithm is the high computational demands it makes. In its simplest and fastest form it runs in worst-case time $O(m^2n)$ on a network with m edges and n vertices, or $O(n^3)$ on a sparse graph (one for which m scales with n in the limit of large n , which covers essentially all networks of current scientific interest, with the possible exception of food webs). With typical computer resources available at the time of writing, this limits the algorithm’s use to networks of a few thousand vertices at most, and substantially less than this for interactive applications. Increasingly however, there is interest in the study of much larger networks; citation and collaboration networks can contain millions of vertices [12, 13] for example, while the world wide web numbers in the billions [14].

In this paper, therefore, we propose a new algorithm for detecting community structure. The algorithm operates on different principles to that of Girvan and Newman (GN) but, as we will show, gives qualitatively similar results. The worst-case running time of the algorithm is $O((m+n)n)$, or $O(n^2)$ on a sparse graph. In practice, it runs to completion on current computers in reason-

able times for networks of up to a million or so vertices, bringing within reach the study of communities in many systems that would previously have been considered intractable.

II. THE ALGORITHM

Our algorithm is based on the idea of modularity. Given any network, the GN community structure algorithm always produces *some* division of the vertices into communities, regardless of whether the network has any natural such division. To test whether a particular division is meaningful we define a quality function or “modularity” Q as follows [6]. Let e_{ij} be the fraction of edges in the network that connect vertices in group i to those in group j , and let $a_i = \sum_j e_{ij}$ [19]. Then

$$Q = \sum_i (e_{ii} - a_i^2) \quad \text{! = nr of comm} \quad (1)$$

is the fraction of edges that fall within communities, minus the expected value of the same quantity if edges fall at random without regard for the community structure. If a particular division gives no more within-community edges than would be expected by random chance we will get $Q = 0$. Values other than 0 indicate deviations from randomness, and in practice values greater than about 0.3 appear to indicate significant community structure. A number of examples are given in Ref. 6.

But this now suggests an alternative approach to finding community structure. If a high value of Q represents a good community division, why not simply optimize Q over all possible divisions to find the best one? By doing this, we can avoid the iterative removal of edges and cut straight to the chase. The problem is that true optimization of Q is very costly. The number of ways to divide n vertices into g non-empty groups is given by the Stirling number of the second kind $S_n^{(g)}$, and hence the number of distinct community divisions is $\sum_{g=1}^n S_n^{(g)}$. This sum is not known in closed form, but we observe that $S_n^{(1)} + S_n^{(2)} = 2^{n-1}$ for all $n > 1$, so that the sum must increase at least exponentially in n . To carry out

an exhaustive search of all possible divisions for the optimal value of Q would therefore take at least an exponential amount of time, and is in practice infeasible for systems larger than twenty or thirty vertices. Various approximate optimization methods are available: simulated annealing, genetic algorithms, and so forth. Here we consider a scheme based on a standard “greedy” optimization algorithm, which appears to perform well.

Our algorithm falls in the general category of agglomerative hierarchical clustering methods [15, 16]. Starting with a state in which each vertex is the sole member of one of n communities, we repeatedly join communities together in pairs, choosing at each step the join that results in the greatest increase (or smallest decrease) in Q . The progress of the algorithm can be represented as a “dendrogram,” a tree that shows the order of the joins (see Fig. 2, for an example). Cuts through this dendrogram at different levels give divisions of the network into larger or smaller numbers of communities and, as with the GN algorithm, we can select the best cut by looking for the maximal value of Q .

Since the joining of a pair of communities between which there are no edges at all can never result in an increase in Q , we need only consider those pairs between which there are edges, of which there will at any time be at most m , where m is again the number of edges in the graph. The change in Q upon joining two communities is given by $\Delta Q = e_{ij} + e_{ji} - 2a_i a_j = 2(e_{ij} - a_i a_j)$, which can clearly be calculated in constant time. Following a join, some of the matrix elements e_{ij} must be updated by adding together the rows and columns corresponding to the joined communities, which takes worst-case time $O(n)$. Thus each step of the algorithm takes worst-case time $O(m + n)$. There are a maximum of $n - 1$ join operations necessary to construct the complete dendrogram and hence the entire algorithm runs in time $O((m + n)n)$, or $O(n^2)$ on a sparse graph. The algorithm has the added advantage of calculating the value of Q as it goes along, making it especially simple to find the optimal community structure.

It is worth noting that our algorithm can be generalized trivially to weighted networks in which each edge has a numeric strength associated with it, by making the initial values of the matrix elements e_{ij} equal to those strengths, rather than just zero or one; otherwise the algorithm is as above and has the same running time. The networks studied in this paper however are all unweighted.

III. APPLICATIONS

As a first example of the working of our algorithm, we have generated using a computer a large number of random graphs with known community structure, which we then run through the algorithm to quantify its performance. Each graph consists of $n = 128$ vertices divided into four groups of 32. Each vertex has on average z_{in}

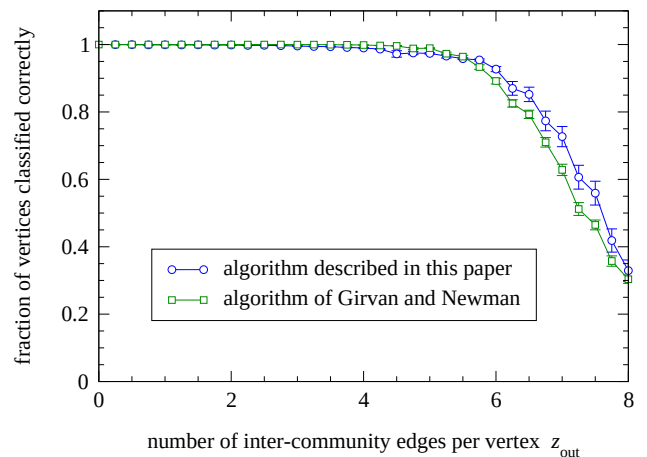


FIG. 1: The fraction of vertices correctly identified by our algorithms in the computer-generated graphs described in the text. The two curves show results for the new algorithm (circles) and for the algorithm of Girvan and Newman [5] (squares). Each point is an average over 100 graphs.

edges connecting it to members of the same group and z_{out} edges to members of other groups, with z_{in} and z_{out} chosen such that the total expected degree $z_{in} + z_{out} = 16$, in this case. As z_{out} is increased from small values, the resulting graphs pose greater and greater challenges to the community-finding algorithm. In Fig. 1 we show the fraction of vertices correctly assigned to the four communities by the algorithm as a function of z_{out} . As the figure shows, the algorithm performs well, correctly identifying more than 90% of vertices for values of $z_{out} \lesssim 6$. Only when z_{out} approaches the value 8 at which the number of within- and between-community edges per vertex is the same does the algorithm begin to fail. On the same plot we also show the performance of the GN algorithm and, as we can see, that algorithm performs slightly but measurably better for smaller values of z_{out} . For example, for $z_{out} = 5$ our new algorithm correctly identifies an average of 97.4(2)% of vertices, while the older algorithm correctly identifies 98.9(1)%. Both, however, clearly perform well.

Interestingly for higher values of z_{out} our new algorithm performs better than the older one, and we have come across a few real-world networks in which this is the case also. Normally, however, the GN algorithm seems to have the edge, and this should come as no great surprise. Our new algorithm bases its decisions on purely local information about individual communities, while the GN algorithm uses non-local information about the entire network—information derived from betweenness scores. Since community structure is itself fundamentally a non-local quantity, it seems reasonable that one can do a better job of finding that structure if one has non-local information at one’s disposal.

For systems small enough that the GN algorithm is computationally tractable, therefore, we see no reason

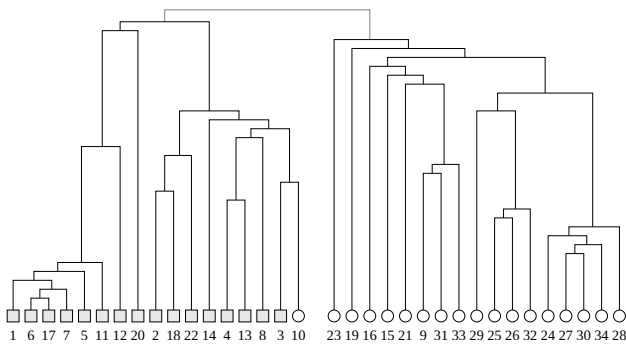


FIG. 2: Dendrogram of the communities found by our algorithm in the “karate club” network of Zachary [5, 17]. The shapes of the vertices represent the two groups into which the club split as the result of an internal dispute.

not to continue using it—it appears to give the best results. For systems too large to make use of this approach, however, our new algorithm gives useful community structure information with comparatively little effort.

We have applied our algorithm to a variety of real-world networks also. We have looked, for example, at the “karate club” network studied in [5], which represents friendships between 34 members of a club at a US university, as recorded over a two-year period by Zachary [17]. During the course of the study, the club split into two groups as a result of a dispute within the organization, and the members of one group left to start their own club. In Fig. 2 we show the dendrogram derived by feeding the friendship network into our algorithm. The peak modularity is $Q = 0.381$ and corresponds to a split into two groups of 17, as shown in the figure. The shapes of the vertices represent the alignments of the club members following the split and, as we can see, the division found by the algorithm corresponds almost perfectly to these alignments; only one vertex, number 10, is classified wrongly. The GN algorithm performs similarly on this task, but not better—it also finds the split but classifies one vertex wrongly (although a different one, vertex 3). In other tests, we find that our algorithm also successfully detects the main two-way division of the dolphin social network of Lusseau [6, 18], and the division between black and white musicians in the jazz network of Gleiser and Danon [11].

As a demonstration of how our algorithm can sometimes miss some of the structure in a network, we take another example from Ref. 5, a network representing the schedule of games between American college football teams in a single season. Because the teams are divided into groups or “conferences,” with intra-conference games being more frequent than inter-conference games, we have a reasonable idea ahead of time about what communities our algorithm should find. The dendrogram generated by the algorithm is shown in Fig. 3, and has an optimal modularity of $Q = 0.546$, which is a little shy

of the value 0.601 for the best split reported in [5]. As the dendrogram reveals, the algorithm finds six communities. Some of them correspond to single conferences, but most correspond to two or more. The GN algorithm, by contrast, finds all eleven conferences, as well as accurately identifying independent teams that belong to no conference. Nonetheless, it is clear that the new algorithm is quite capable of picking out useful community structure from the network, and of course it is much the faster algorithm. On the author’s personal computer the algorithm ran to completion in an unmeasurably small time—less than a hundredth of a second. The algorithm of Girvan and Newman took a little over a second.

A time difference of this magnitude will not present a big problem in most practical situations, but performance rapidly becomes an issue when we look at larger networks; we expect the ratio of running times to increase with the number of vertices. Thus, for example, in applying our algorithm to the 1275-node network of jazz musician collaborations mentioned above, we found that it runs to completion in about one second of CPU time. The GN algorithm by contrast takes more than three hours to reach very similar results.

As an example of an analysis made possible by the speed of the new algorithm, we have looked at a network of collaborations between physicists as documented by papers posted on the widely-used Physics E-print Archive at arxiv.org. The network is an updated version of the one described in Ref. 13, in which scientists are considered connected if they have coauthored one or more papers posted on the archive. We analyze only the largest component of the network, which contains $n = 56\,276$ scientists in all branches of physics covered by the archive. Since two vertices that are unconnected by any path are never put in the same community by our algorithm, the small fraction of vertices that are not part of the largest component can safely be assumed to be in separate communities in the sense of our algorithm. Our algorithm takes 42 minutes to find the full community structure. Our best estimates indicate that the GN algorithm would take somewhere between three and five years to complete its version of the same calculation.

The analysis reveals that the network in question consists of about 600 communities, with a high peak modularity of $Q = 0.713$, indicating strong community structure in the physics world. Four of the communities found are large, containing between them 77% of all the vertices, while the others are small—see Fig. 4, left panel. The four large communities correspond closely to subject subareas: one to astrophysics, one to high-energy physics, and two to condensed matter physics. Thus there appears to be a strong correlation between the structure found by our algorithm and the community divisions perceived by human observers. It is precisely correlation of this kind that makes community structure analysis a useful tool in understanding the behavior of networked systems.

We can repeat the analysis with any of the subcom-

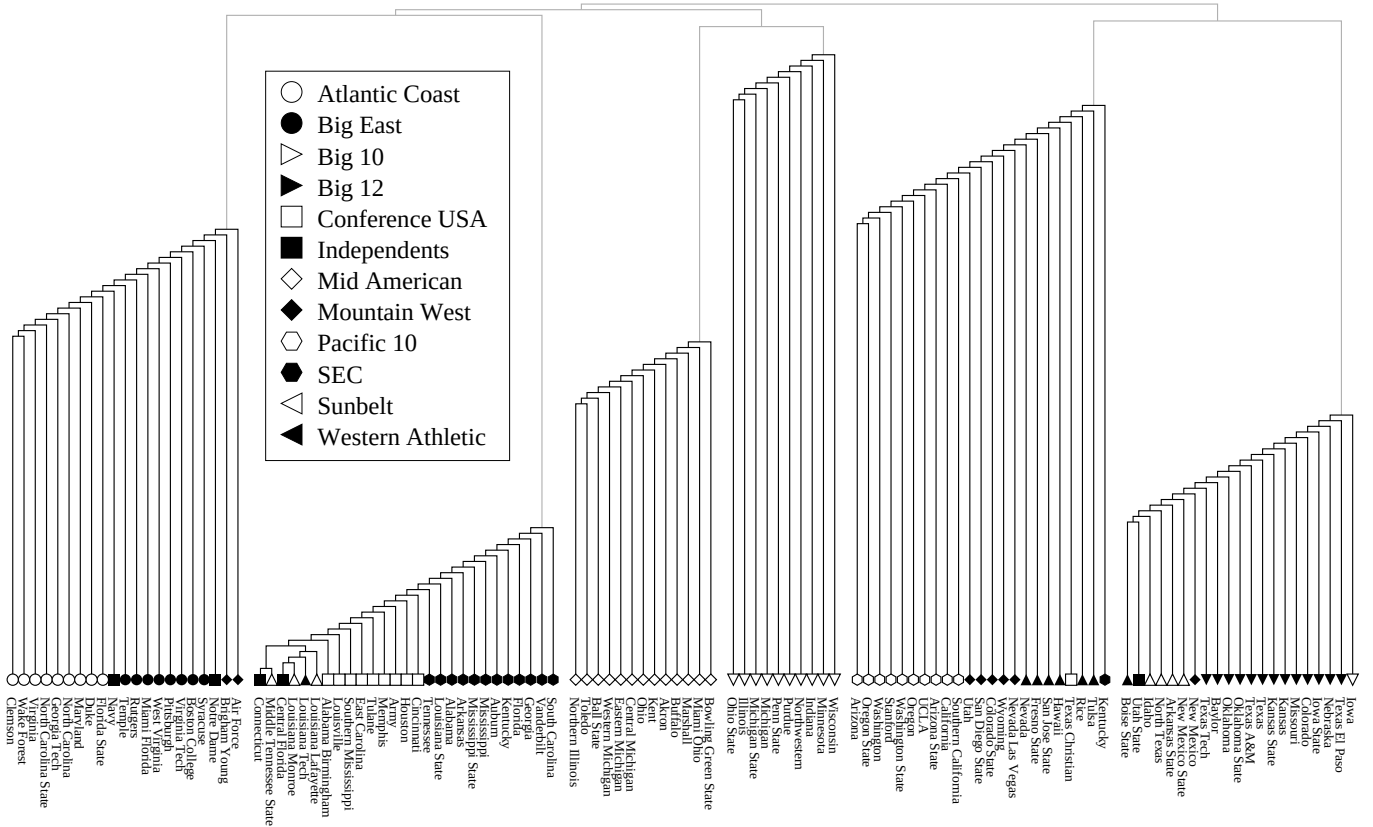


FIG. 3: Dendrogram of the communities found in the college football network described in the text. The real-world communities—conferences—are denoted by the different shapes as indicated in the legend.

munities to observe how they break up. For example, feeding the smaller of the two condensed-matter groups through the algorithm again, we find an even stronger peak modularity of $Q = 0.807$ —the strongest we have yet observed in any network—corresponding to a split into about a hundred communities of all sizes (Fig. 4, center panel). The sizes appear roughly to have a power-law distribution with exponent about -2 [20]. Narrowing our focus still further to the particular one of these communities that contains the present author, we find the structure shown in the right panel of Fig. 4. Feeding this one last time through the algorithm, it breaks apart into communities that correspond closely to individual institutional research groups, the author’s group appearing in the corner of the figure, highlighted by the dashed box. One could pursue this line of analysis further, identifying individual groups, iteratively breaking them down, and looking for example at the patterns of collaboration between them, but we leave this for later studies.

IV. CONCLUSIONS

In this paper we have described a new algorithm for extracting community structure from networks, which has

a considerable speed advantage over previous algorithms, running to completion in time that scales as the square of the network size. This allows us to study much larger systems than has previously been possible. Among other examples, we have applied the algorithm to a network of collaborations between more than fifty thousand physicists, and found that the resulting community structure corresponds closely to the traditional divisions between specialties and research groups in the field.

We believe that our method will not only allow for the extension of community structure analysis to some of the very large networks that are now being studied for the first time, but also provides a useful tool for visualizing and understand the structure of these networks, whose daunting size has hitherto made many of their structural properties obscure.

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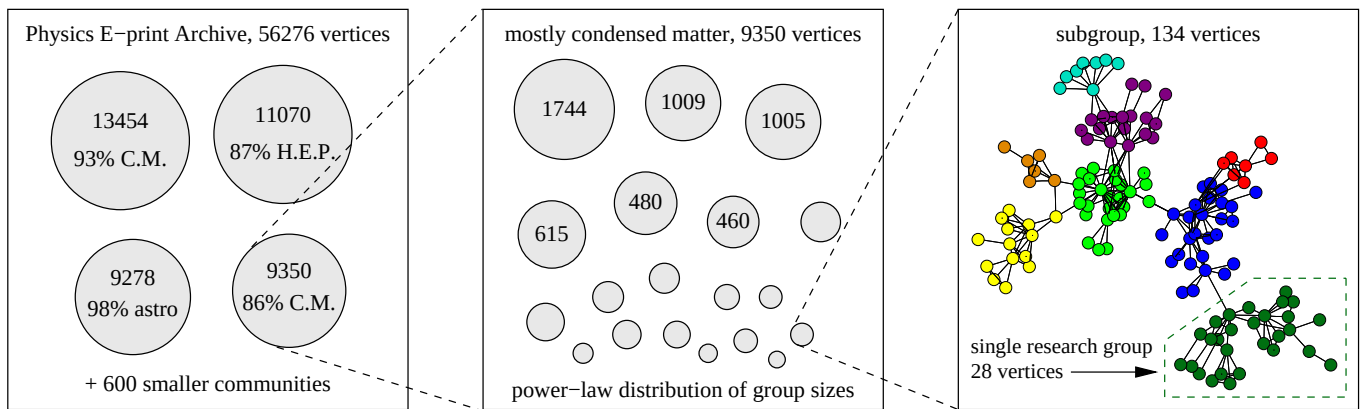


FIG. 4: Left panel: Community structure in the collaboration network of physicists. The graph breaks down into four large groups, each composed primarily to physicists of one specialty, as shown. Specialties are determined by the subsection(s) of the e-print archive in which individuals post papers: “C.M.” indicates condensed matter; “H.E.P.” high-energy physics including theory, phenomenology, and nuclear physics; “astro” indicates astrophysics. Middle panel: one of the condensed matter communities is further broken down by the algorithm, revealing an approximate power-law distribution of community sizes. Right panel: one of these smaller communities is further analyzed to reveal individual research groups (different shades), one of which (in dashed box) is the author’s own.

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 - [19] As discussed in [6], each edge should contribute only to e_{ij} once, either above or below the diagonal, but not both. Alternatively, and more elegantly, one can split the contribution of each edge half-and-half between e_{ij} and e_{ji} , except for those edges that join a group to itself, whose contribution belongs entirely to the single diagonal element e_{ii} for the group in question.
 - [20] This power law is of a different kind to the one observed in an email network by Guimerà *et al.* [9]. They studied the histogram of community sizes over all levels of the dendrogram; we are looking only at the single level corresponding to the maximum value of Q .